

posterior density form:

$$p(\theta|x) = \frac{p(x|\theta)\pi(\theta)}{\int_{-\infty}^{\infty} p(x|t)\pi(t) dt}$$

What each part means

- $p(\theta|x)$ $p(\theta \mid x)p(\theta|x)$: posterior density of parameter θ after seeing data x
- $p(x|\theta)$ $p(x \mid \theta)p(x|\theta)$: likelihood
- $\pi(\theta)$ $\pi(\theta)\pi(\theta)$: prior density
- The denominator is the **normalizing constant**, ensuring the posterior integrates to 1

Key idea

The integral replaces the discrete sum:

$$P(A_i|B)=\frac{P(B|A_i)P(A_i)}{\sum_j P(B|A_j)P(A_j)}P(A_i|B)=\sum_j \frac{P(B|A_i)P(A_i)}{P(B|A_j)P(A_j)}P(A_j|B)P(A_j)$$

$$p(A_i|B) = \frac{P(B|A_i)P(A_i)}{\sum_j P(B|A_j)P(A_j)}$$

So in continuous spaces:

- **sum → integral**
- **probability mass → density**